A1.1 Water

A1 Unity and diversity
Molecules

SL

Understandings:

- A1.1.1—Water as the medium for life
- A1.1.2—Hydrogen bonds as a consequence of the polar covalent bonds within water molecules
- A1.1.3—Cohesion of water molecules due to hydrogen bonding and consequences for organisms
- A1.1.4—Adhesion of water to materials that are polar or charged and impacts for organisms
- A1.1.5—Solvent properties of water linked to its role as a medium for metabolism and for transport in plants and animals
- A1.1.6—Physical properties of water and the consequences for animals in aquatic habitats

Key words:

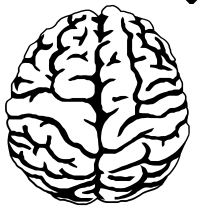
Solvent
Solute
Solution
Hydrogen bonds
Polar molecule
Cohesion
Transpiration
Surface tension
Specific heat capacity
Latent heat of vaporisation/ evaporation
Adhesion
Capillary action
Hydrophilic
Hydrophobic
Buoyancy
Viscosity
Thermal conductivity

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



Guiding questions

What physical and chemical properties of water make it essential for life?

What are the challenges and opportunities of water as a habitat?

A1.1.1—Water as the medium for life



What percentage of the human body is water?

What percentage of the human body is water?

Around 60%

Water is essential for life.

Has properties which allow it to be:

- A universal solvent
- Provide structural support
- Involved in essential chemical reactions

The first life on planet earth probably started in water!

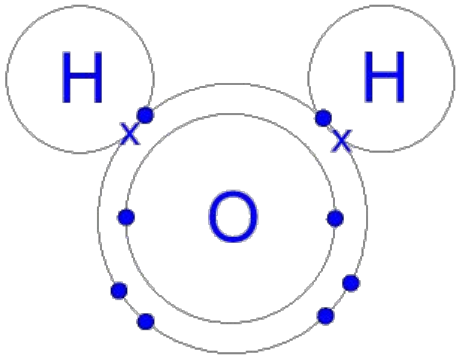


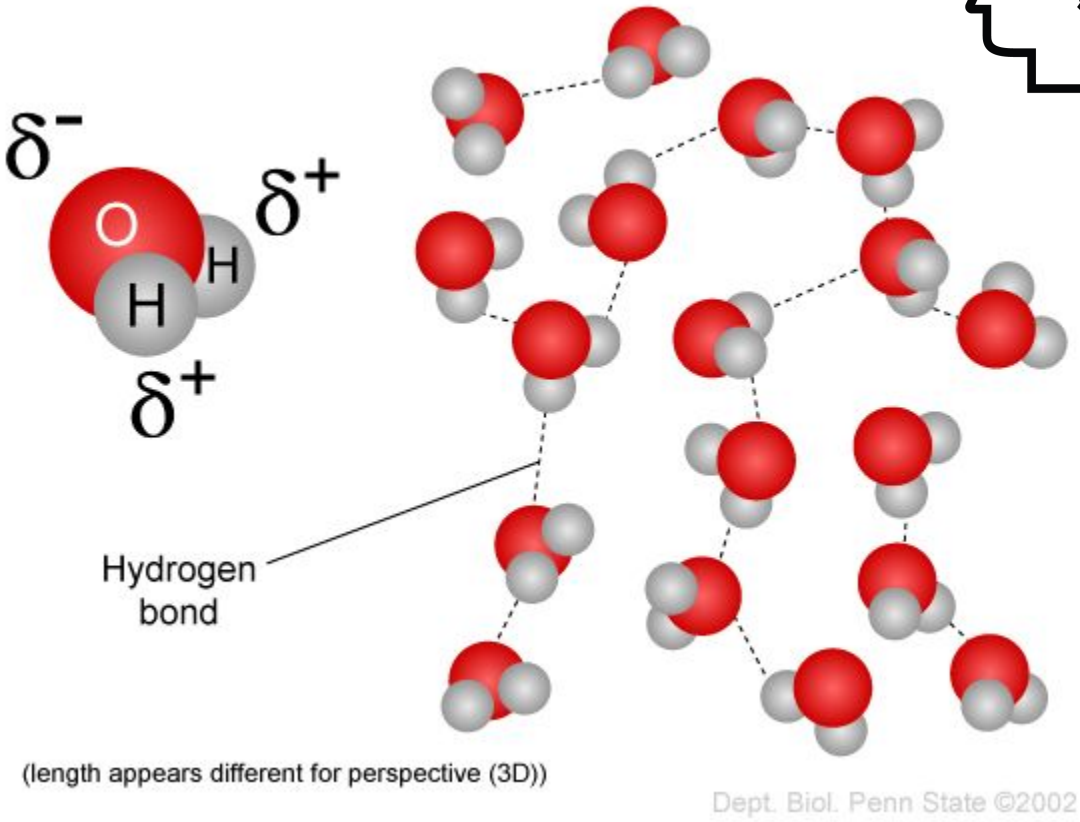
Why does a wet paper towel stick to the window?

A1.1.2—Hydrogen bonds as a consequence of the polar covalent bonds within water molecules

Polar molecule

Can you explain why the hydrogen atoms are together on one side?







Ping pong balls and stickers -

Place four stickers (electrons) as far apart from each other as possible on the surface of the ping pong ball (oxygen atom)

Imagine that a hydrogen atom is bonded to two of these

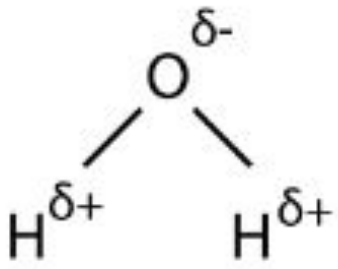
As the oxygen is a larger atom (and has more protons in its nucleus) it exerts more of a force of attraction on the shared electrons.

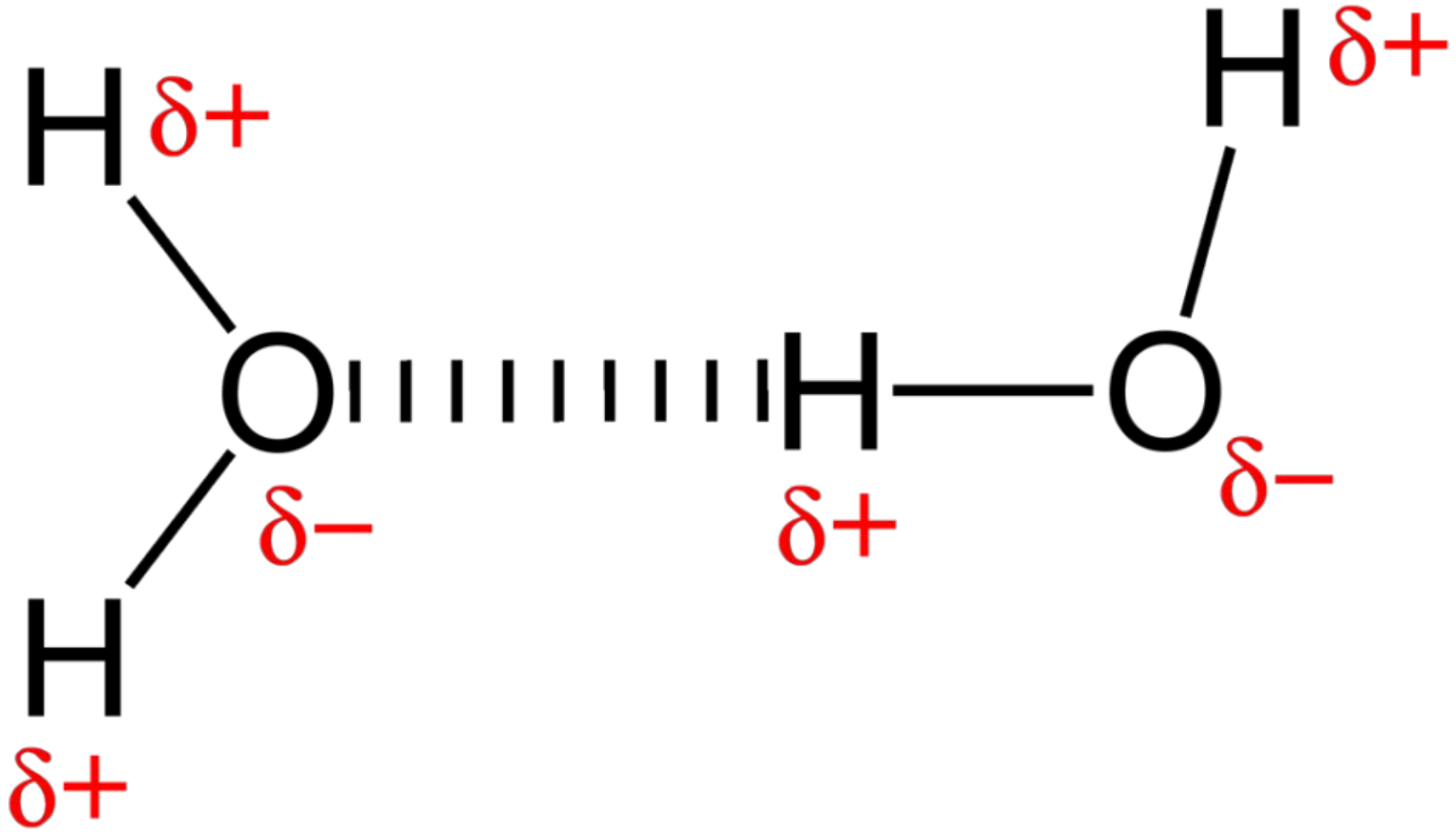
It has a small relative negative charge

As the hydrogen is a smaller atom (and has fewer protons in its nucleus) it exerts less of a force of attraction on the shared electrons

It has a small relative positive charge

You need to be able to draw this diagram!




H
δ+
H
δ+
O
H
O
δ−
δ+
δ−
H
δ+

Water is said to be **polar** - it has a negative and a positive pole.

The polar nature of water gives it many important properties

A1.1.3—Cohesion of water molecules due to hydrogen bonding and consequences for organisms

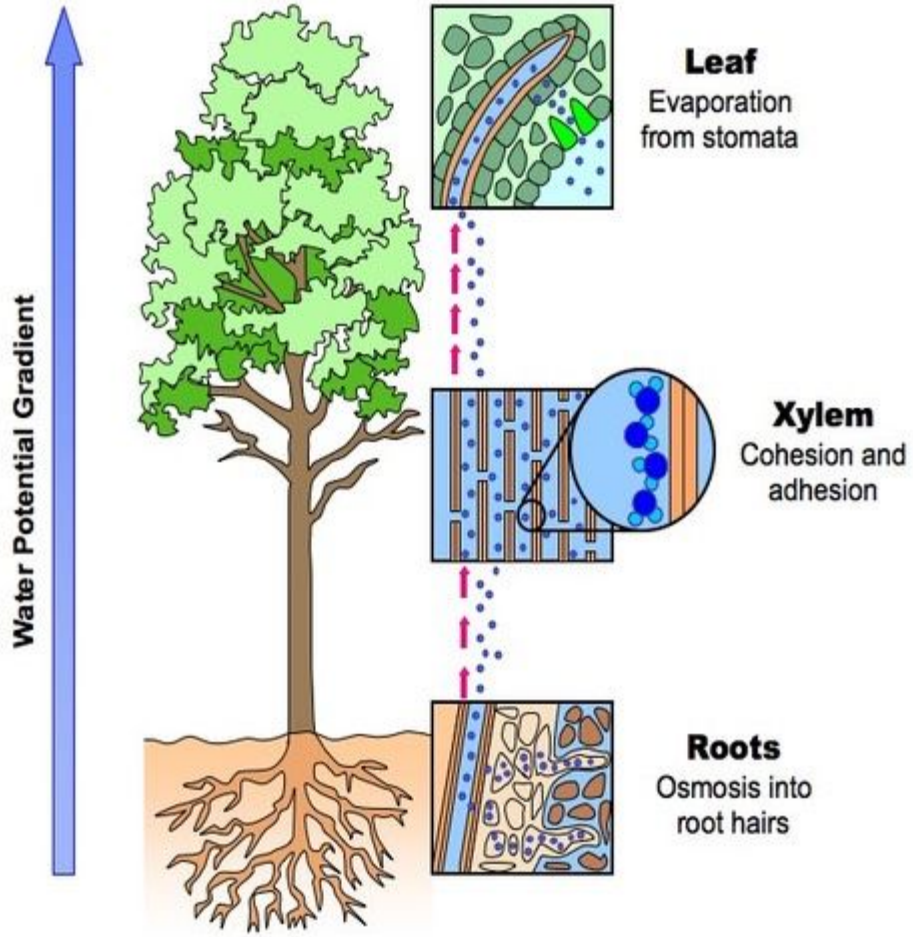
Cohesion



Cohesion - When water molecules are attracted to other water molecules via hydrogen bonding.

Can you think of any biological processes in plants that are only possible due to the cohesion of water?

Transpiration

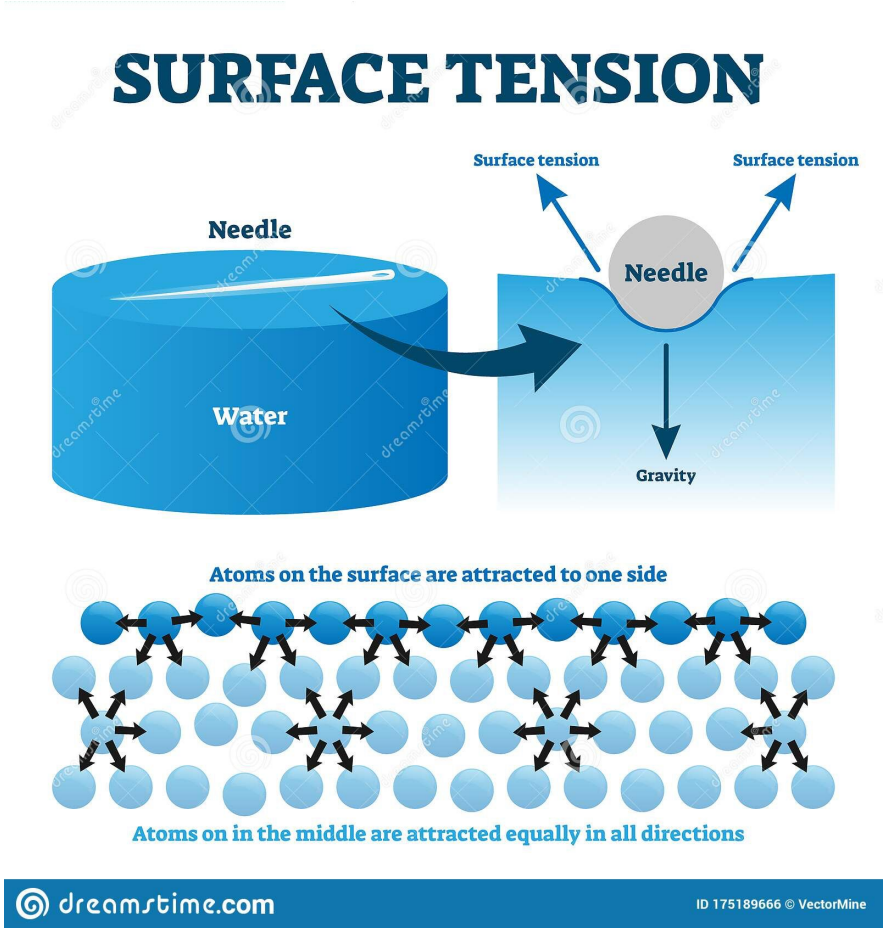




Explain how some insects are able to walk on water.

Surface tension







Why is water such a hospitable environment to live?

(why is water usually neither too hot nor too cold to live in?)

High specific heat capacity

Ability of molecule to absorb energy

Water has highest specific heat capacity of any liquid.

Requires lot of energy to increase temperature

- Amount of energy required to raise temperature of 1g of water by 1°C = 1 calorie = 4.184 Joules

Maintains internal and external environment with large fluctuations of temperature (habitat/ body)

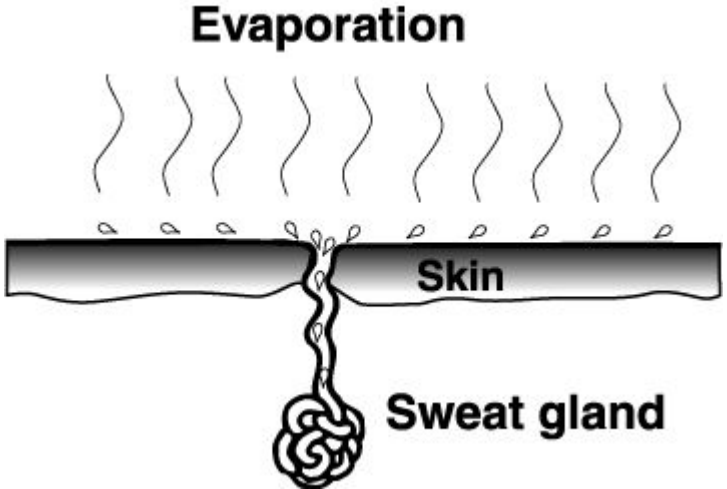
https://lab.concord.org/embeddable.html#interactives/sam/intermolecular-attractions/6-hydrogen-bonds-a-special-type-of-attraction.json

High temp, more movement, energy into breaking hydrogen bonds
Low temp, less movement, energy into forming hydrogen bonds

Latent heat of vaporisation/ evaporation

The hydrogen bonds between the water molecules also mean that it takes more energy for water to evaporate.

This property of water makes at an excellent coolant, e.g. sweating.



A1.1.4—Adhesion of water to materials that are polar or charged and impacts for organisms

Adhesion

Adhesion - When water molecules are attracted to other polar substances via hydrogen bonding.

Adhesion is also essential in transpiration.

The water is attracted to the walls of the xylem - preventing the column from falling down.

Why does water travel against gravity up a piece of tissue?

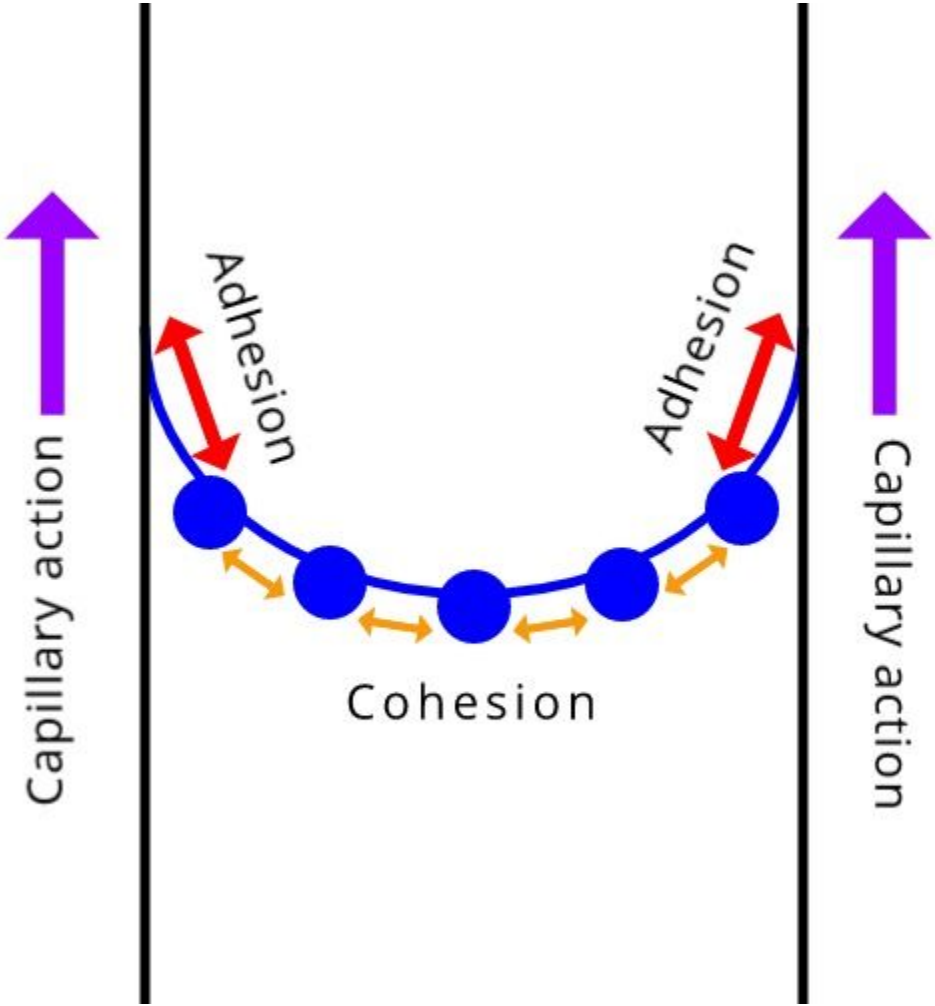


Capillary action

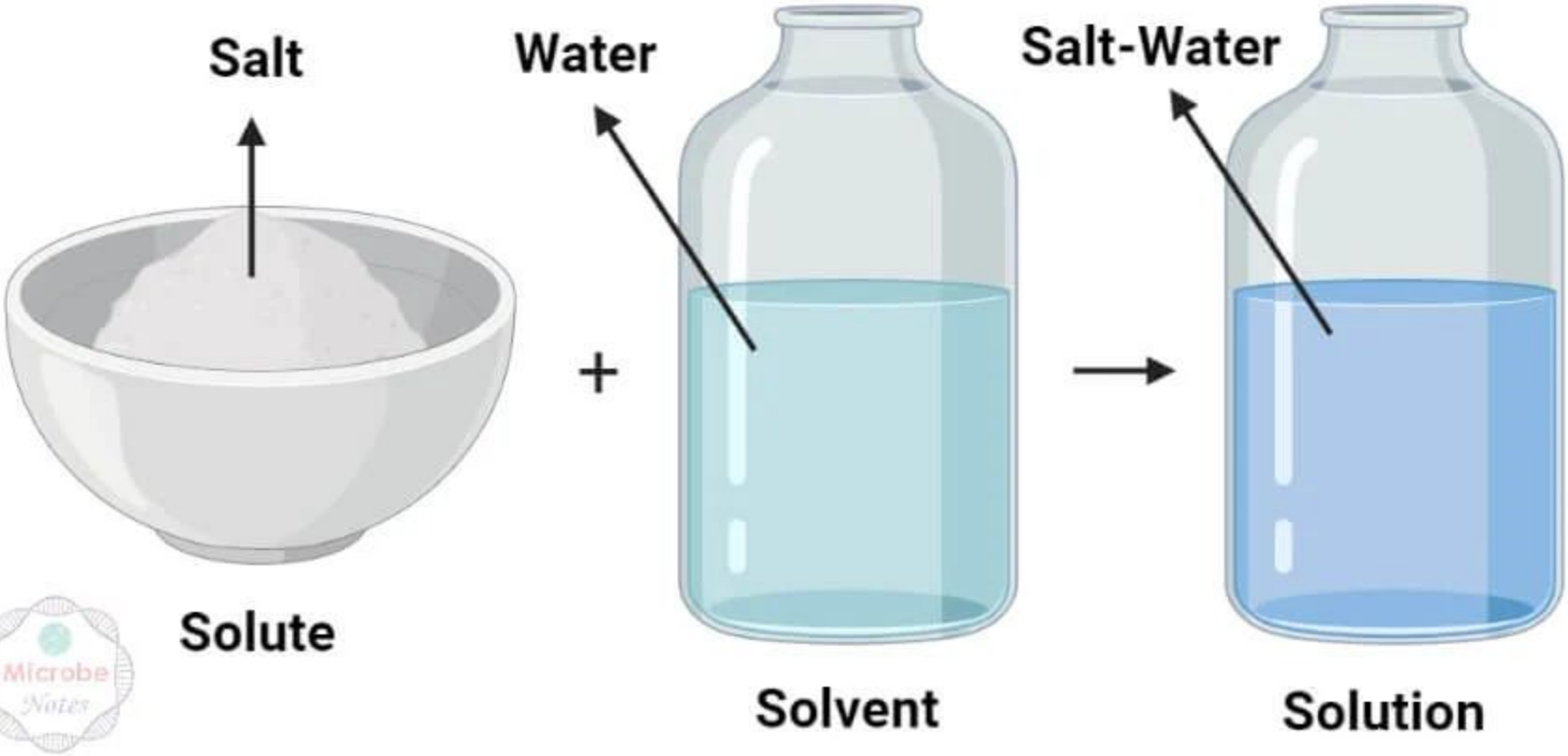
The cellulose of the paper is polar

The walls of the tissue attract the water molecules.

Adhesion (and capillary action) allows water to travel up through soil to the roots of plants and also around cells via the cellulose cell wall



A1.1.5—Solvent properties of water linked to its role as a medium for metabolism and for transport in plants and animals

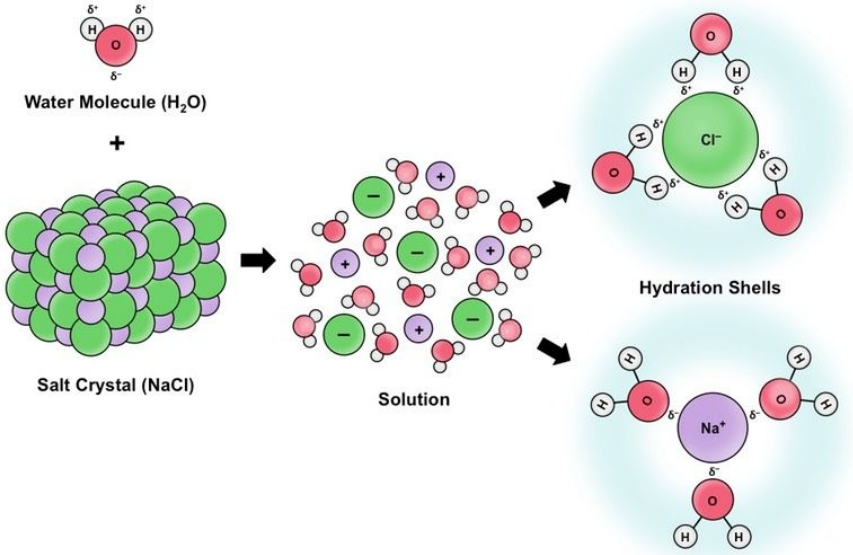

Salt
Water
Salt-Water
+
Solute
Solvent
Solution
Microbe Notes

Polar solvent

Mixes readily with other polar substances and ions

Polar water molecules squeeze in between charges dissolving solute

What are the terms for substances that do and do not associate with water?



Polar solvent

Mixes readily with other polar substances and ions

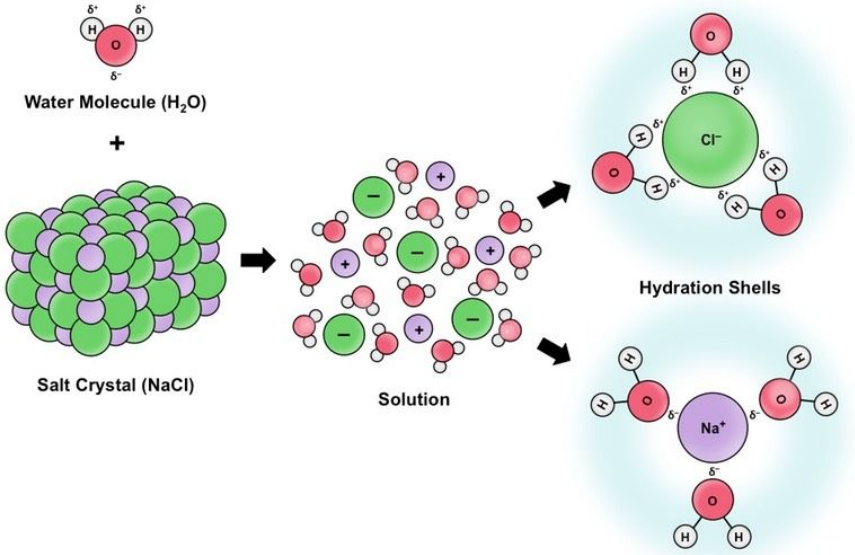
Polar water molecules squeeze in between charges dissolving solute

Hydrophilic

associates with water

Hydrophobic

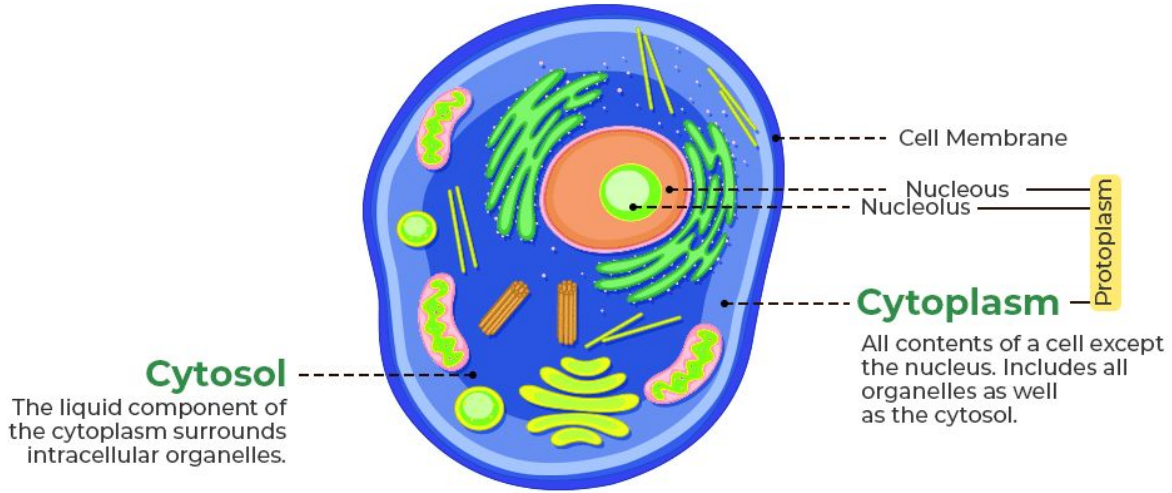
does not associate with water



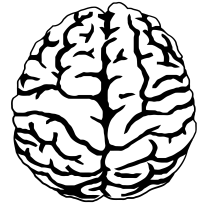
Metabolism

Enzymes and substrates are found dissolved in the cytoplasm of cells.

The water allows metabolic reactions to take place



Transport



Water is the medium for transport in large organisms such as plants and animals.

What are the transport systems in plants and what do they transport?

What is the main transport system in humans and what does it transport?

Transport

Water is the medium for transport in large organisms such as plants and animals.

What are the transport systems in plants and what do they transport?

Xylem - water and mineral ions

Phloem - sugars and amino acids

What is the main transport system in humans and what does it transport?

Blood - cells (RBCs, WBCs), nutrients and waste products

A1.1.6—Physical properties of water and the consequences for animals in aquatic habitats

Buoyancy - aquatic organisms have a similar density to water meaning they can swim through easily

Viscosity - water creates resistance to organisms swimming through it. Salt water is more viscous than fresh-water, which is more viscous than air

Thermal conductivity - water conducts heat very well. Warm blooded aquatic animals need a layer of insulating fat to stop them from losing all their body heat

Specific heat capacity - the temperature of the water is relatively stable over a wide range of external temperatures



Using all that you have learned about the properties of water, discuss the different advantages and challenges posed by the two habitats of the emperor penguin (*Aptenodytes forsteri*) - in water and on land



Can you...



Describe how hydrogen bonds are formed between polar molecules

Describe the difference between cohesion and adhesion

List some consequences of cohesion of water to organisms

List a consequence of adhesion of water to organisms

Explain why water is known as the universal solvent

Describe how water is used in transport in plants and animals

List some properties of water which are relevant to aquatic animals